

Introduction to Game Theory: Quiz 1

30 minutes, 25 points

Closed book. One handwritten A4 sheet (both sides) is allowed. No calculator. Justify all answers.

Question 1 (6 points). Consider the following game, where player 1 chooses a row and player 2 chooses a column.

	L	C	R
U	3, 2	1, 3	0, 1
M	2, 2	2, 2	1, 1
D	1, 3	1, 1	4, 0

Eliminate dominated strategies iteratively. Which strategies survive? Show each elimination step.

Question 2 (12 points). Consider the game

	L	R
T	4, 2	0, 0
B	0, 0	2, 4

- (a) (4 points) Find all pure-strategy Nash equilibria.
- (b) (6 points) Find the mixed-strategy Nash equilibrium in which both players randomize.
- (c) (2 points) Compute player 1's equilibrium payoff.

Question 3 (7 points). Suppose that in a finite strategic game every player i has a strictly dominant strategy s_i^* . Prove that $s^* = (s_1^*, \dots, s_n^*)$ is a Nash equilibrium and that it is the only Nash equilibrium in pure strategies.